# Supplementary Materials for: Competing feedback channels drive phase transitions in collective emotion

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## Supplementary Contents

# Guide to this Supplementary Materials

Core results are reported in the main text; this supplement provides supporting methodological details and auxiliary analyses. Sections S1–S2 expand the *Materials and Methods*, and Sections S3–S6 provide supporting figures referenced from the *Results*. The main text cites these materials as "Supplementary Materials S#" and "Supplementary Fig. S#".

**Contents overview.**

- **Supplementary Methods**
  - **S1** LLM annotation protocol and validation.
  - **S2** Mean-field derivations for the critical point and Ginzburg–Landau form.
- **Supplementary Results**
  - **S3** Network validation details and order-parameter conventions.
  - **S4** Early-warning signal analysis for critical slowing down.
  - **S5** Parameter effects on the critical boundary.
  - **S6** Empirical analysis details and density diagnostics.

**Quick reference (main text $\leftrightarrow$ supplement).**

| Main text | Supplementary location |
|---|---|
| Methods: LLM annotation | S1 |
| Methods: mean-field derivations | S2 |
| Results: direction–intensity coupling | Fig. S2 (S3) |
| Results: robustness across networks | Figs. S1–S4 (S3) |
| Results: critical slowing down | Figs. S5–S6 (S4) |
| Results: parameter landscape | Figs. S7–S12 (S5) |
| Results: empirical signatures | Figs. S13–S16 (S6) |

**Notation.** Supplementary Table S1 summarizes the main symbols used throughout the paper, together with baseline values and ranges where applicable.

Table S1: Notation and baseline values.

| Symbol | Meaning | Baseline / definition |
|---|---|---|
| **Model state and environment** | | |
| $H, M, L$ | high/medium/low arousal states | — |
| $\rho_H, \rho_M, \rho_L$ | population fractions in $H, M, L$ | — |
| $q$ | polarization direction ($q = \rho_H - \rho_L$) | $q \in [-1, 1]$ |
| $a$ | activity / moderate-buffer depletion ($a = \rho_H + \rho_L = 1 - \rho_M$) | $a \in [0, 1]$; in data $a = X_H + X_L$ |
| $Q$ | observed polarization order parameter | ABM: analog of $q$; data: $Q = X_H - X_L$ |
| $p_{\text{env}}$ | perceived risk probability in the mixed environment | weighted mixture of channel risks (Section S2) |
| **Model parameters** | | |
| $\phi$ | high-arousal threshold | 0.54 |
| $\theta$ | low-arousal threshold | 0.46 |
| $k$ | signals sampled per update (information density) | 50 (baseline); scanned 10–500 |
| $n_m$ | mainstream channel strength | 10 |
| $n_w$ | user-generated channel strength | 5 |
| $n_w/n_m$ | media ratio (user-generated vs. mainstream strength) | 0.5 (baseline); scanned 0.2–2.0 |
| $r$ | channel-balance control parameter | varied in $[0, 1]$ |
| $\beta$ | local coupling strength (ABM) | 0 (baseline); varied up to 0.2 |
| **Simulation settings and diagnostics** | | |
| $N$ | population size | varies by figure |
| $\langle k \rangle$ | network average degree | 50 (unless stated otherwise) |
| $f$ | update fraction per step | 0.1; 1 sweep = $N$ updates = $1/f$ steps |
| $U_4(r; N)$ | Binder cumulant used for finite-size scaling | $1 - \langle Q^4 \rangle / (3\langle Q^2 \rangle^2)$ |
| **Theory quantities** | | |
| $\chi$ | psychological sensitivity of collective response | $\chi = \frac{d\mathcal{S}}{dp}\Big\rvert_{p=1/2}$ |
| $\Gamma(r)$ | feedback gradient of $p_{\text{env}}$ around $q = 0$ | defined in Section S2 |
| $r_c$ | critical point (stability boundary) | defined in Section S2 |
| $\Delta r_c$ | ABM–theory deviation in the estimated boundary | $\Delta r_c = r_c^{\text{ABM}} - r_c^{\text{theory}}$ |
| $\tau$ | relaxation time near criticality | $\tau \sim 1/\lvert r_c - r \rvert$ as $r \to r_c^-$ |
| $\tau_a/\tau_q$ | relaxation-time ratio (activity vs. polarization) | estimated in asymmetric simulations; $\approx 0.70$ (Fig. S2) |
| $\text{AC}_1$ | lag-1 autocorrelation | computed from steady-state $Q(t)$ |
| **Empirical proxies** | | |
| $X_H, X_M, X_L$ | fractions of posts labeled $H/M/L$ within a time window | — |
| $r_{\text{proxy}}$ | user-generated dominance proxy | $n_{\text{wemedia}}/(n_{\text{wemedia}} + n_{\text{mainstream}} + n_{\text{government}})$ |
| $\sigma_Q$ | volatility within a segment | $\sigma_Q = \text{std}(Q)$ |
| $\text{jump}_{q95}$ | jump intensity within a segment | 95th percentile of $\lvert \Delta Q/\Delta t \rvert$ |

# Supplementary Methods

Sections S1–S2 provide additional methodological details for the empirical annotation pipeline and the mean-field derivations.

# S1 LLM annotation protocol

We annotate Weibo posts along two dimensions—emotional arousal and risk content—using `Qwen/Qwen3-8B` [1]. Prompts are structured and enforce a fixed JSON output schema to improve reproducibility.

## S1.1 Annotation setup

We serve the model via a local OpenAI-compatible API. Inference uses temperature = 0.1 and max_tokens = 512.

## S1.2 Label definitions

**Emotional arousal (`emotion_class`).**

- **H**: strong emotional activation (anger, panic/fear, sarcasm/insults, highly emotional accusations).
- **M**: neutral or informational content (calm discussion, explanations, supportive messages).
- **L**: negative but low-intensity emotions (worry, confusion, helplessness, sadness).

**Risk content (`risk_class`).**

- **risk**: conveys negative health impacts of COVID-19/long COVID (symptoms, severity, sequelae, warnings).
- **norisk**: emphasizes recovery/controllability or reassurance, criticizes fear-mongering, or is unrelated to health risks.

## S1.3 Prompt structure and JSON output

The prompt specifies the above criteria, forbids any free-form text beyond a single JSON object, and requests confidence scores in $[0, 1]$. Output fields are: `emotion_class`, `emotion_confidence`, `risk_class`, `risk_confidence`, and a brief `reasoning` string.

## S1.4 Prompt template (English translation)

The annotation prompt is implemented as a system message containing the task definitions and criteria, followed by a user message containing the post text. For readability, we show an English translation of the core prompt template below (the original Chinese prompt is provided in the code repository).

```
[System]
You are a social-media content analyst. Classify the following Weibo post along:
1) emotional arousal (H/M/L) and 2) health-risk content (risk/norisk).
Return ONLY a single JSON object. No markdown, no extra text.

Output fields (JSON):
  emotion_class: "H" | "M" | "L"
  emotion_confidence: float in [0,1]
  risk_class: "risk" | "norisk"
  risk_confidence: float in [0,1]
```

```
  reasoning: <= 20 words

Emotion (emotion_class):
  H: strong emotion (anger, panic/fear, sarcasm/insults, emotional accusations)
  M: neutral/informational (calm discussion, explanations, supportive messages)
  L: negative but low-intensity (worry, confusion, helplessness, sadness)

Risk (risk_class):
  risk: conveys negative health impacts of COVID-19/long COVID (symptoms, warnings)
  norisk: reassurance/recovery/controllability, critiques fear-mongering, or unrelated

If the post is too short or contains only hashtags with no substantive content:
  default emotion_class=M and risk_class=norisk with confidence=0.5.

[User]
Post: {post_text}

[Output]
Return ONLY a valid JSON object matching the schema below.
```

### S1.5 JSON schema

The model output is constrained to a fixed JSON object with the following fields:

```
{
  "emotion_class": "H" | "M" | "L",
  "emotion_confidence": 0.0 - 1.0,
  "risk_class": "risk" | "norisk",
  "risk_confidence": 0.0 - 1.0,
  "reasoning": "<=20 words>"
}
```

We discard any response that does not parse as JSON or violates the allowed label set.

### S1.6 Validation

We compared LLM-assigned labels to manual ratings on a validation subset ($n = 5{,}000$) randomly sampled from the corpus after de-duplication by post identifier (`mid`). To reduce anchoring effects, manual labels were collected in a blind annotation sheet that does not include LLM outputs, and the LLM labels were joined only at scoring time. Exact-match accuracy was 85% for arousal labels (Cohen's $\kappa = 0.79$) and 87% for risk labels ($\kappa = 0.82$). Fine-grained metrics and confusion matrices are reported in Supplementary Tables S2–S5.

### S1.7 Fine-grained performance metrics

For each class, we compute precision and recall from the confusion matrix and report 95% confidence intervals using Wilson score intervals. F1-score intervals are obtained by propagating the corresponding precision/recall bounds through $F_1 = 2PR/(P + R)$.

Table S2: Fine-grained performance metrics for three-class emotional arousal classification (H/M/L; $n = 5{,}000$).

| Category | Precision [95% CI] | Recall [95% CI] | F1-score [95% CI] |
|---|---|---|---|
| H (high arousal) | 0.86 [0.84, 0.88] | 0.84 [0.82, 0.86] | 0.85 [0.83, 0.87] |
| M (medium arousal) | 0.83 [0.81, 0.85] | 0.87 [0.85, 0.89] | 0.85 [0.83, 0.87] |
| L (low arousal) | 0.86 [0.84, 0.88] | 0.84 [0.82, 0.86] | 0.85 [0.83, 0.87] |
| Overall | Accuracy = 0.85, Cohen's $\kappa = 0.79$ | | |

Table S3: Confusion matrix for emotional arousal classification (H/M/L; $n = 5{,}000$).

| | Predicted | | |
|---|---|---|---|
| Actual | H | M | L |
| H | **1412** | 185 | 83 |
| M | 124 | **1451** | 95 |
| L | 106 | 112 | **1432** |

Table S4: Fine-grained performance metrics for risk-content classification (`risk`/`norisk`; $n = 5{,}000$).

| Category | Precision [95% CI] | Recall [95% CI] | F1-score [95% CI] |
|---|---|---|---|
| `norisk` | 0.89 [0.88, 0.90] | 0.92 [0.91, 0.93] | 0.90 [0.89, 0.91] |
| `risk` | 0.81 [0.79, 0.83] | 0.75 [0.73, 0.77] | 0.78 [0.76, 0.80] |
| Overall | Accuracy = 0.87, Cohen's $\kappa = 0.82$ | | |

Table S5: Confusion matrix for risk-content classification (`risk`/`norisk`; $n = 5{,}000$).

| | Predicted | |
|---|---|---|
| Actual | `norisk` | `risk` |
| `norisk` | **3220** | 280 |
| `risk` | 370 | **1130** |

# S2 Mean-field derivations

This note provides the derivations omitted from the main text for the mean-field critical point and the Ginzburg–Landau form. Notation and baseline values are summarized in Supplementary Table S1.

## S2.1 Mean-field update and collective response

Under well-mixed assumptions, each update samples $k$ independent risk signals from the environment, each labeled "risk" with probability $p_{\text{env}}$. Let $S \sim \text{Binomial}(k, p_{\text{env}})$ and $\hat{p} = S/k$ be the perceived risk fraction. Given thresholds $\theta < \phi$, an updated individual moves to $H$ if $\hat{p} \geq \phi$, to $L$

if $\hat{p} \le \theta$, and otherwise remains in $M$. Therefore,

$$P_H(p_{\text{env}}) = \Pr(\hat{p} \ge \phi) = \sum_{s=\lceil k\phi \rceil}^{k} \binom{k}{s} p_{\text{env}}^{s} (1 - p_{\text{env}})^{k-s}, \tag{1}$$

$$P_L(p_{\text{env}}) = \Pr(\hat{p} \le \theta) = \sum_{s=0}^{\lfloor k\theta \rfloor} \binom{k}{s} p_{\text{env}}^{s} (1 - p_{\text{env}})^{k-s}. \tag{2}$$

The expected polarization direction after an update is the difference between high- and low-arousal outcomes,

$$\mathcal{S}(p_{\text{env}}) = P_H(p_{\text{env}}) - P_L(p_{\text{env}}). \tag{3}$$

With asynchronous updates and a continuous-time approximation, the mean-field dynamics take the form

$$\frac{dq}{dt} = -q + \mathcal{S}(p_{\text{env}}), \tag{4}$$

which is the mean-field form used in the main text.

### S2.2 Linear stability and analytic critical point

In the symmetric regime,

$$p^{\text{main}}(q) = \frac{1-q}{2}, \qquad p^{\text{we}}(q) = \frac{1+q}{2}. \tag{5}$$

The environmental risk is the weighted average of the two channels,

$$p_{\text{env}}(q; r) = \frac{(1-r)n_m p^{\text{main}}(q) + r n_w p^{\text{we}}(q)}{(1-r)n_m + r n_w}, \tag{6}$$

where $n_m$ and $n_w$ are baseline channel strengths and $r \in [0, 1]$ controls their relative weight. Expanding around $q = 0$ yields

$$p_{\text{env}}(q; r) = \frac{1}{2} + \Gamma(r)\, q + O(q^3), \tag{7}$$

where the feedback gradient is

$$\Gamma(r) = \left.\frac{\partial p_{\text{env}}}{\partial q}\right|_{q=0} = \frac{r n_w - (1-r)n_m}{2\left((1-r)n_m + r n_w\right)}. \tag{8}$$

Next, expand the collective response around neutrality $p = 1/2$:

$$\mathcal{S}(p) = \mathcal{S}(1/2) + \chi \left(p - \frac{1}{2}\right) + O\!\left(\left(p - \frac{1}{2}\right)^3\right), \qquad \chi = \left.\frac{d\mathcal{S}}{dp}\right|_{p=1/2}. \tag{9}$$

Under symmetry, $\mathcal{S}(1/2) = 0$, so substituting $p_{\text{env}}(q; r)$ into the mean-field dynamics gives, to leading order,

$$\frac{dq}{dt} = (-1 + \chi\Gamma)\, q + O(q^3). \tag{10}$$

Therefore the neutral fixed point $q = 0$ loses stability when the linear restoring coefficient changes sign:

$$\chi\Gamma(r_c) = 1. \tag{11}$$

Solving for $r$ yields the closed-form critical point reported in the main text:

$$r_c = \frac{n_m(\chi+2)}{n_m(\chi+2)+n_w(\chi-2)}. \tag{12}$$

When $\chi \le 2$, the denominator does not admit a critical point in $[0,1]$, consistent with the absence of a transition for weak psychological sensitivity.

### S2.3 Ginzburg–Landau form and noise term

Keeping the next nonlinearity in the expansion yields a cubic term. Writing

$$\mathcal{S}(p) = \chi\left(p-\frac{1}{2}\right) + \frac{\mathcal{S}^{(3)}(1/2)}{6}\left(p-\frac{1}{2}\right)^3 + O\left(\left(p-\frac{1}{2}\right)^5\right), \tag{13}$$

and using $p_{\mathrm{env}} - 1/2 \approx \Gamma q$, we obtain

$$\frac{dq}{dt} = (-1+\chi\Gamma)\,q + \frac{\mathcal{S}^{(3)}(1/2)}{6}\Gamma^3 q^3 + \eta(t) + \cdots = -\alpha q - u q^3 + \eta(t) + \cdots, \tag{14}$$

where $\alpha = 1-\chi\Gamma$ and $u = -\mathcal{S}^{(3)}(1/2)\Gamma^3/6$. For the parameter ranges considered, $u > 0$, so the dynamics can be written in the effective-potential form

$$\frac{dq}{dt} = -\frac{\partial V_{\mathrm{eff}}(q)}{\partial q} + \eta(t), \qquad V_{\mathrm{eff}}(q) = \frac{1}{2}\alpha q^2 + \frac{u}{4}q^4, \tag{15}$$

which gives a pitchfork bifurcation when $\alpha$ changes sign.

The noise term $\eta(t)$ represents stochastic fluctuations induced by finite population size and discrete updates. In the large-$N$ limit, these fluctuations can be approximated as Gaussian white noise with an amplitude that scales as $1/\sqrt{N}$, implying a variance scaling as $1/N$.

# Supplementary Results

Sections S3–S6 provide supporting figures and auxiliary analyses that complement the main-text Results and document conventions used throughout.

## S3 Network validation details and order-parameter conventions

This section clarifies the order-parameter conventions used to visualize the transition, reports the activity–polarization relaxation-time comparison under asymmetry, and documents finite-size diagnostics used to locate the transition in sparse networks.

### S3.1 Order parameter under symmetric branch selection

In the symmetric regime, different realizations can select either the $+q^*$ or $-q^*$ branch with approximately equal probability. As a result, the signed mean $\langle Q\rangle$ cancels even when the bifurcation exists, while $|Q|$ (or sign-aligned $Q$) recovers the transition curve used in the main text.

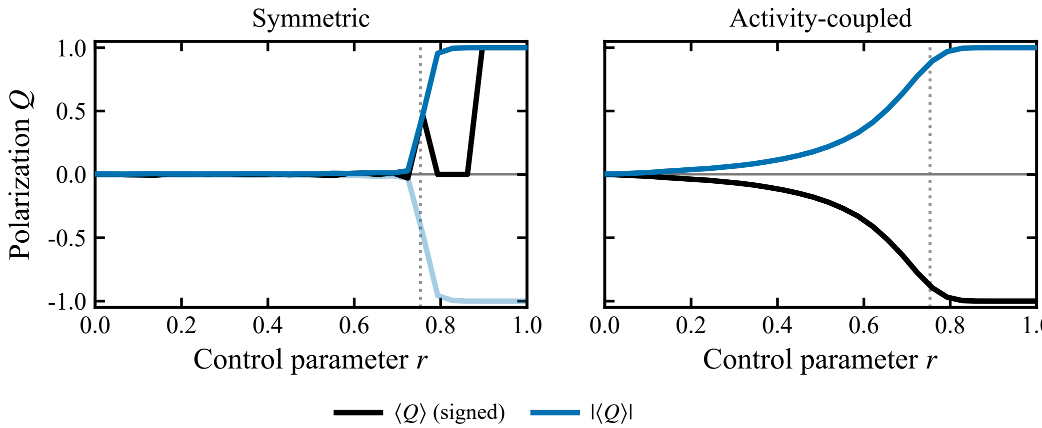


Figure S1: Signed mean $\langle Q \rangle$ versus magnitude $|Q|$ under symmetric branch selection. In the symmetric regime, trajectories select $\pm q^*$ with equal probability, so $\langle Q \rangle$ cancels while $|Q|$ reveals the transition.

### S3.2 Activity is not a passive slow variable under asymmetry

The main text interprets activity as an observable proxy for the depletion of the moderate buffer. In activity-coupled asymmetric simulations, activity often relaxes faster than polarization, implying that $a$ cannot be treated as a passive slow variable and should be tracked jointly with $q$ when testing dynamical signatures and interpreting empirical associations.

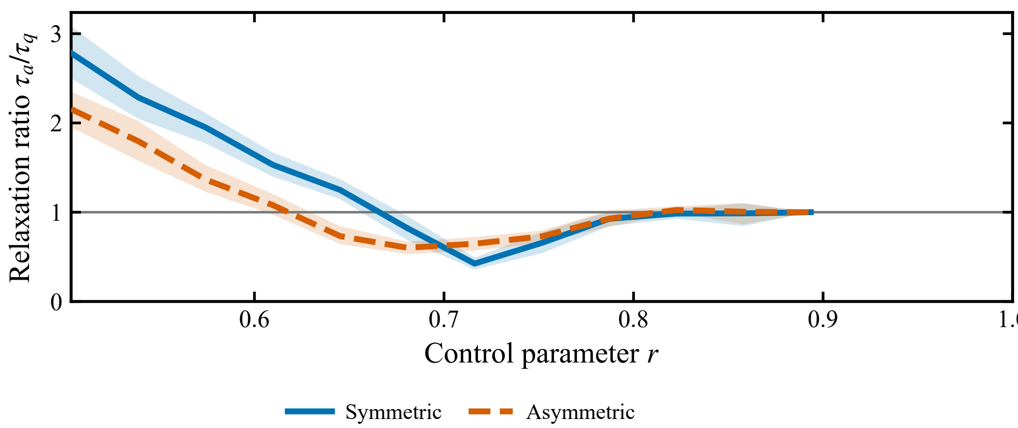


Figure S2: Relaxation-time ratio between activity and polarization under activity-coupled asymmetry. Estimated relaxation times satisfy $\tau_a/\tau_q < 1$ (baseline mean $\approx 0.70$), so activity relaxes faster than polarization.

### S3.3 Finite-size diagnostics for locating the transition

To estimate the transition point in finite sparse networks, we use Binder cumulant crossings because they provide a stable finite-size signature. In contrast, susceptibility peaks can shift non-monotonically with system size and can exhibit pseudo-peaks, making them less reliable for locating $r_c$ in finite simulations.

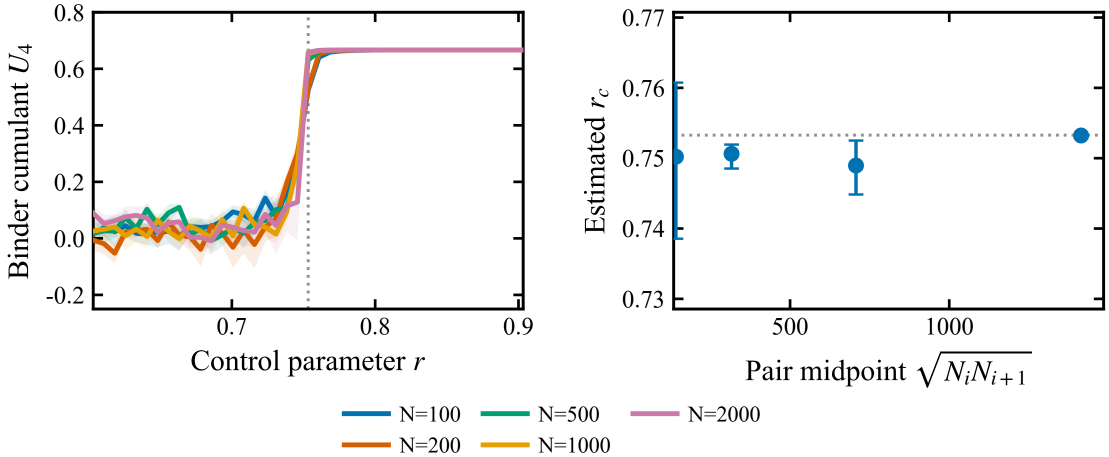


Figure S3: Finite-size scaling via Binder cumulant crossings. Curves $U_4(r; N)$ for multiple system sizes cross near the critical region, providing a stable finite-size estimator of the transition.

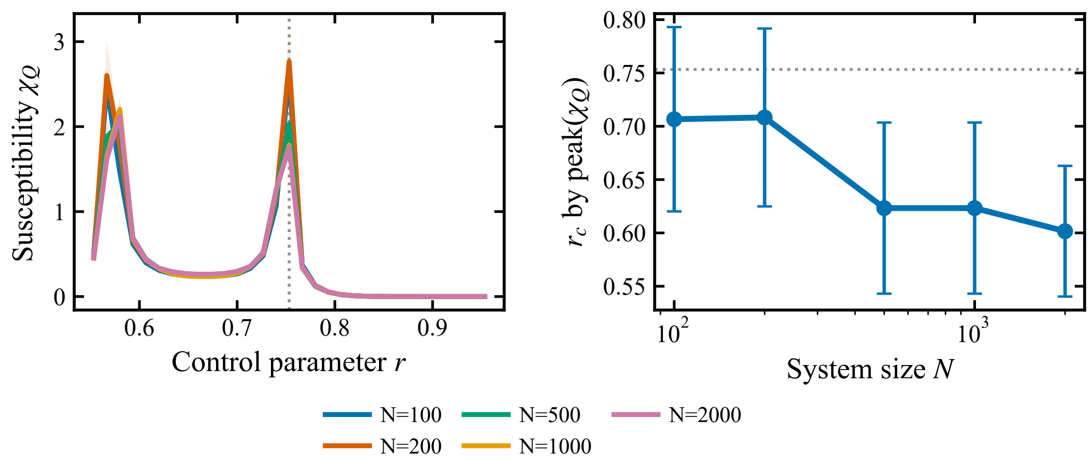


Figure S4: Susceptibility peak locations across system sizes. Peaks can shift non-monotonically and exhibit finite-size artifacts compared with Binder crossings.

# S4 Early-warning signal analysis for critical slowing down

This section reports additional early-warning indicators for critical slowing down beyond the main-text panels.

## S4.1 Early-warning statistics across the control parameter

Approaching the critical point from below, the restoring force weakens and the dynamics become more persistent. This manifests as rising short-lag autocorrelation and variance, which are standard early-warning statistics in the critical-transition literature.

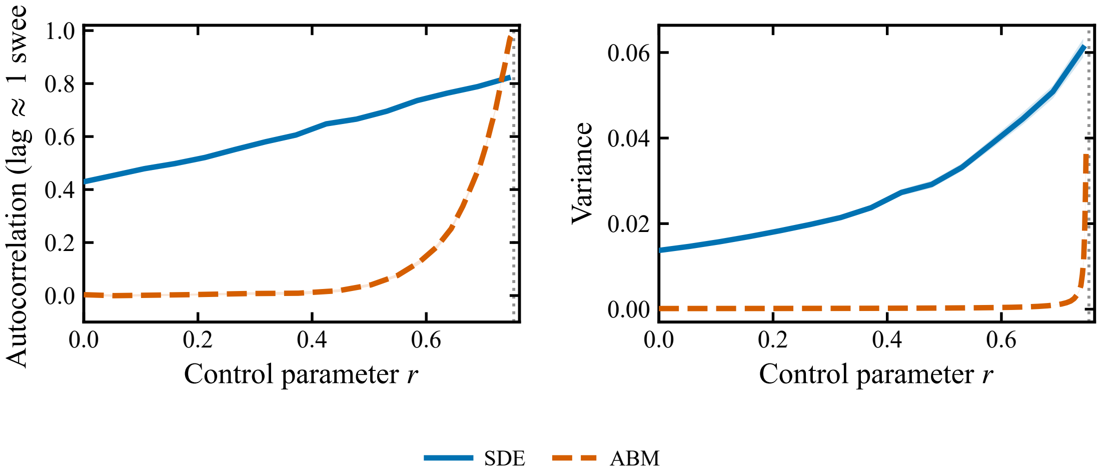


Figure S5: Early-warning indicators across the control parameter. Short-lag autocorrelation and variance increase as $r$ approaches $r_c$ from below.

### S4.2 Robustness across lag choices

To confirm that the persistence increase is not an artifact of choosing a particular lag, we compute autocorrelation at multiple lags. The qualitative rise near criticality is consistent across lags.

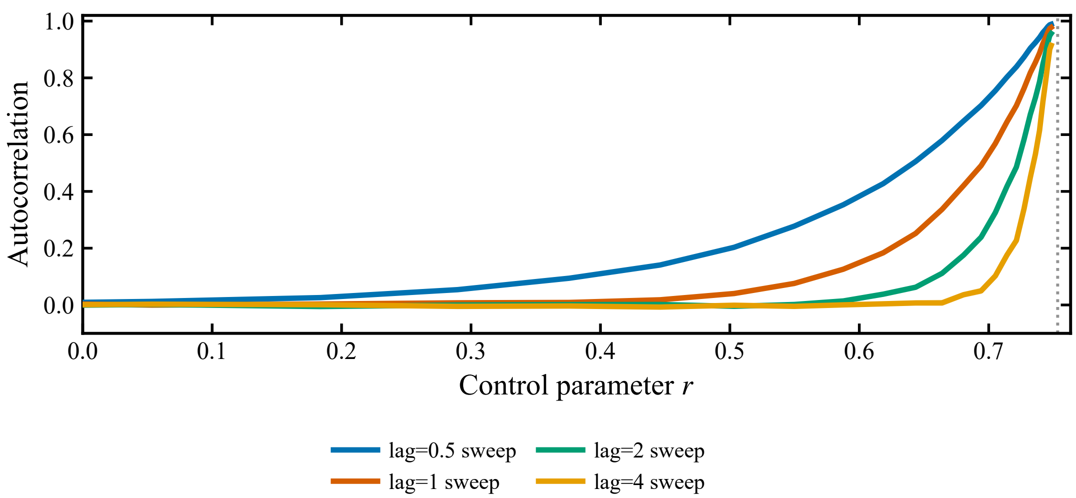


Figure S6: Multi-lag autocorrelation in ABM trajectories. Autocorrelation increases near criticality across multiple lags.

## S5 Parameter effects on the critical boundary

This section provides supporting figures for the parameter-landscape results, clarifying how structural parameters move the critical boundary and documenting checks related to symmetry and finite-size effects.

**Parameter scans and estimators.** The main-text landscape figures evaluate the mean-field critical point on a dense $(\phi, \theta)$ grid over the feasible region $\theta < 0.5 < \phi$. Simulation-based sweeps vary information density $k \in \{10, 20, 50, 100, 200, 500\}$, media ratio $n_w/n_m \in [0.1, 2.0]$ (40 values), and local coupling $\beta \in \{0, 0.02, 0.05, 0.1, 0.2\}$. For each configuration we sample $r$ on a 201-point grid in $[0, 1]$ and estimate $r_c$ from the maximum slope of the median $|Q|(r)$ across seeds, with 95% confidence intervals obtained by bootstrap resampling.

## S5.1 Symmetry validation

The analytic critical point is derived in the symmetric regime. Along the symmetry constraint, the theoretical constructions satisfy neutrality conditions used in the derivation, supporting the internal consistency of the landscape map.

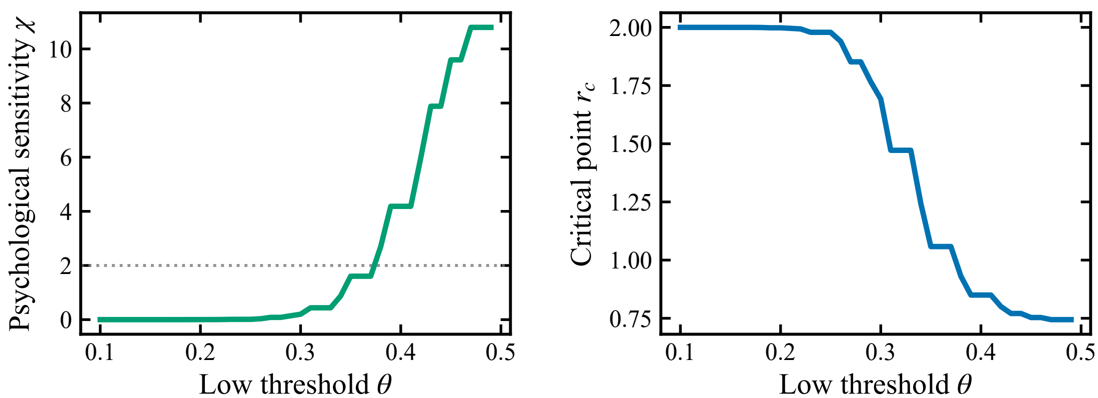


Figure S7: Symmetry diagnostics in the symmetric regime along the analytic constraint in $(\phi, \theta)$.

## S5.2 Decomposing the information-density effect

The main text reports a non-monotonic $k$ effect on the estimated transition point. We separate this into two components: how $k$ changes the psychological sensitivity $\chi$, and how the induced $\chi(k)$ maps to $r_c(k)$.

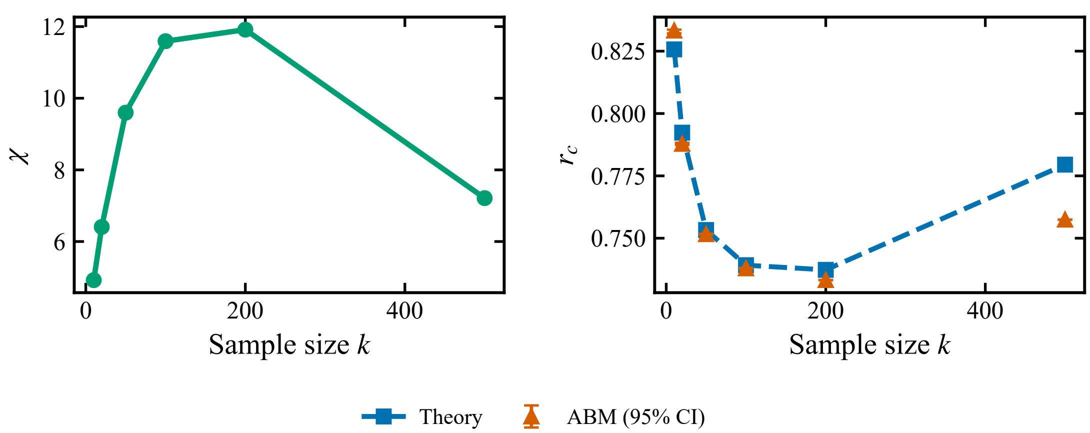


Figure S8: Decomposing the information-density effect into sensitivity $\chi(k)$ and the induced shift in $r_c(k)$.

## S5.3 Finite-size closure at high information density

At very large information density ($k = 500$), discrete-threshold effects and finite-size fluctuations can bias naive estimators. Local scanning and finite-size extrapolation reconcile the apparent deviation with the analytic prediction.

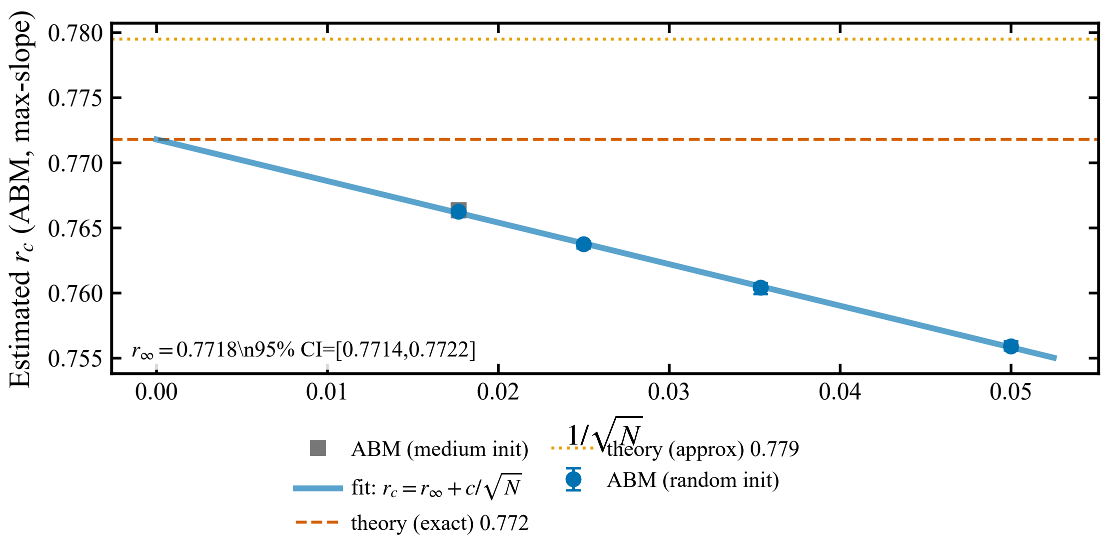


Figure S9: Finite-size closure for $k = 500$ via local scanning and extrapolation.

## S5.4 Local coupling and branch-selection bias

Beyond the well-mixed benchmark, local coupling can introduce reinforcement that both shifts the effective boundary and biases branch selection. These effects help interpret why topology and local interactions matter for proximity to criticality.

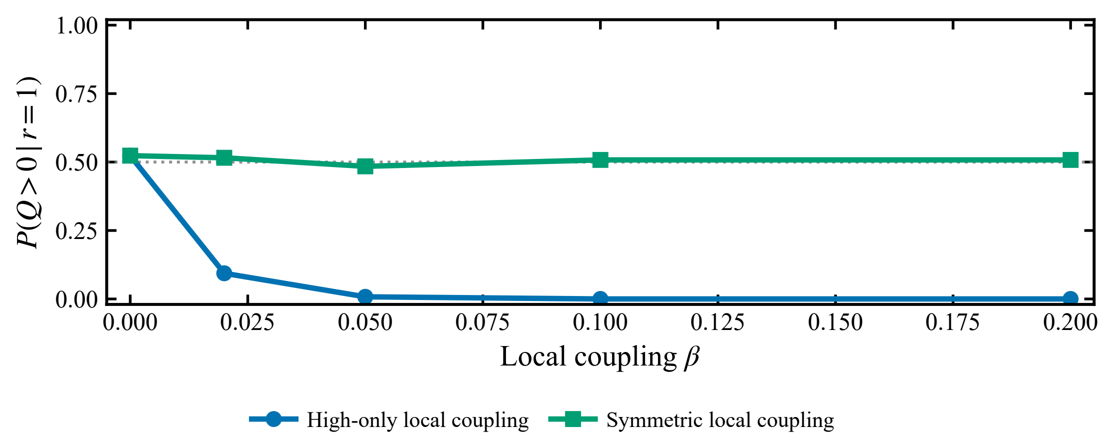


Figure S10: Branch-selection bias induced by local coupling $\beta > 0$.

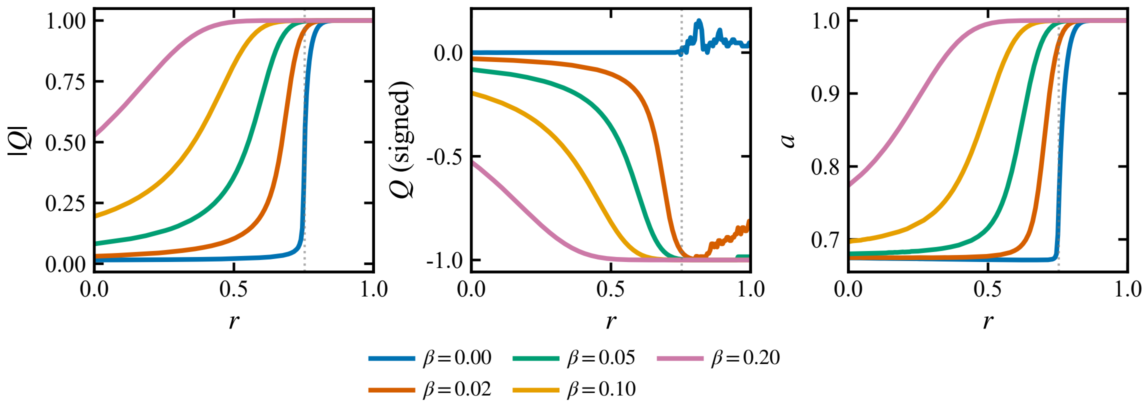


Figure S11: Joint $(q, a)$ behavior under local coupling, showing how polarization and activity co-vary as the boundary shifts.

## S5.5 Parameter importance and interaction

To summarize the magnitude of boundary shifts across parameters, we compare the full scan range $\Delta r_c = \max(r_c) - \min(r_c)$ (normalized by the baseline $r_c$). We also show a theory-level interaction: higher sensitivity $\chi(k)$ amplifies the leverage of media ecology on $r_c$.

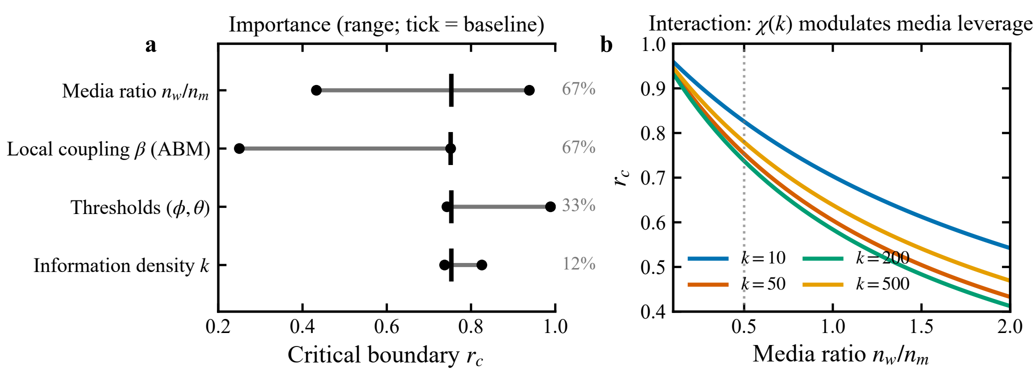


Figure S12: Parameter-importance summary and theory interaction. (a) Range of $r_c$ across parameter sweeps (line: min–max; tick: baseline), annotated by $\Delta r_c/r_{c,\text{base}}$. (b) Theory prediction of $r_c(n_w/n_m)$ for representative $k$ values (vertical dotted line: baseline ratio).

# S6 Empirical analysis details and density diagnostics

This section provides supporting empirical figures, focusing on sampling-density constraints and on diagnostics aligned to the theoretical signatures.

**Empirical subsets.** We use the same subset naming as in the main text: the *pooled dataset* (all eligible segments) and the *high-density subset* (segments with sufficient within-window coverage for stable $r_{\text{proxy}}$ estimation). When shown for context only, we also report a topic-focused *single-topic subset.*

## S6.1 Time-series overview and segmentation rationale

We summarize the window-level proxies used in the main text and highlight missingness patterns. This motivates segment-based aggregation and density-aware diagnostics used in subsequent panels.

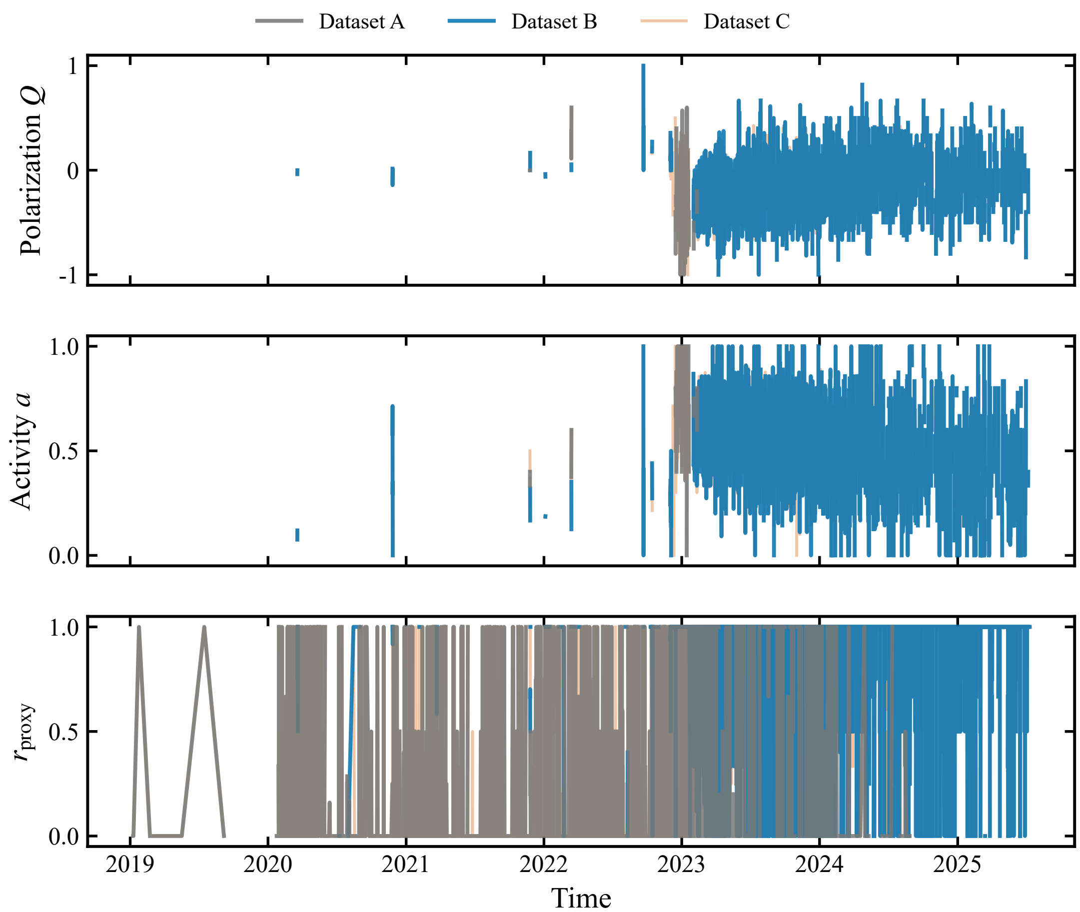


Figure S13: Time-series overview and sampling density. We show the 4-hour aggregates used to compute window-level proxies $(Q, a, r_{\text{proxy}})$, highlighting non-uniform sampling density and missing windows that motivate segment-based aggregation in the main-text analyses.

## S6.2 Scatter diagnostics for the two empirical signatures

To connect directly to the theoretical mechanism, we evaluate two segment-level associations under the primary 4-hour aggregation: elevated activity predicting larger polarization jumps, and increased user-generated dominance predicting higher polarization volatility. Density stratification is shown to diagnose coverage-driven confounds. The main text reports the density-controlled H2 estimate under 12-hour aggregation; the 4-hour diagnostics here provide additional context across datasets and missingness patterns.

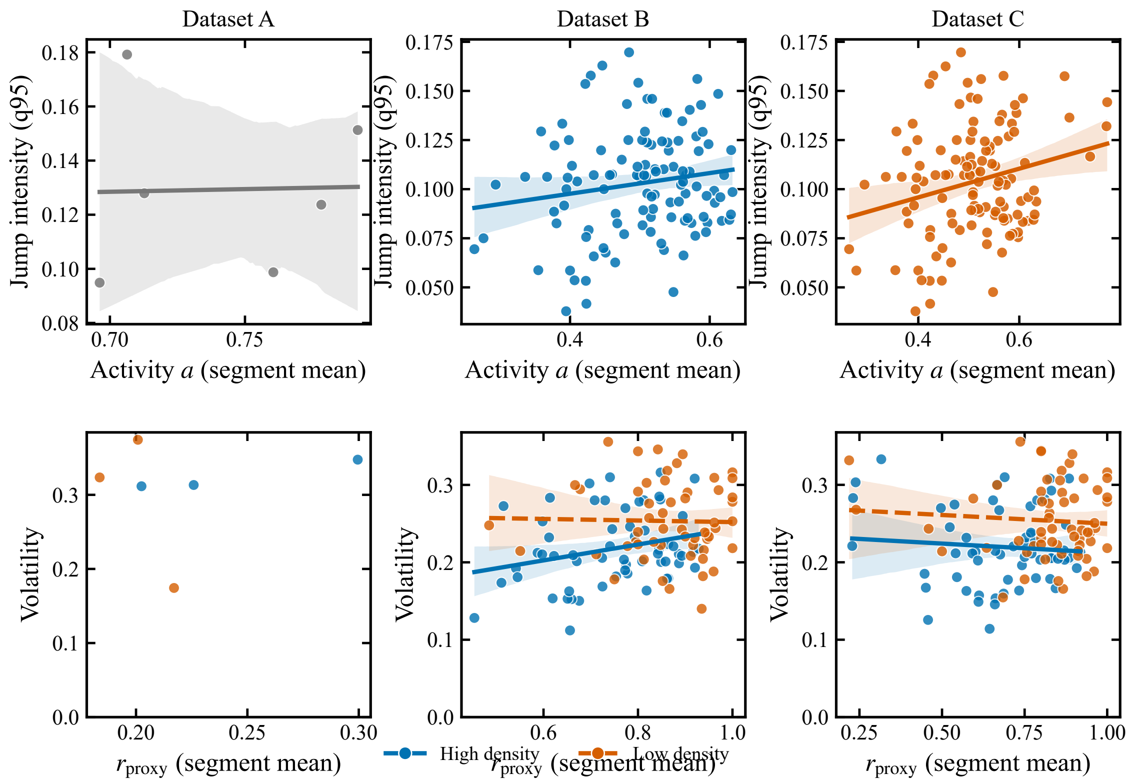


Figure S14: Scatter diagnostics for empirical signatures under the primary 4-hour aggregation. We show segment-level associations for the activity–jump and media-dominance–volatility tests, including density-stratified views used to diagnose sampling-coverage confounds.

## S6.3 Event-aligned early-warning analysis under realistic missingness

We additionally assess early-warning behavior by aligning autocorrelation/variance indicators to jump events. Continuity requirements can make high-frequency aggregation infeasible in sparse datasets, which limits the strength of event-aligned tests.

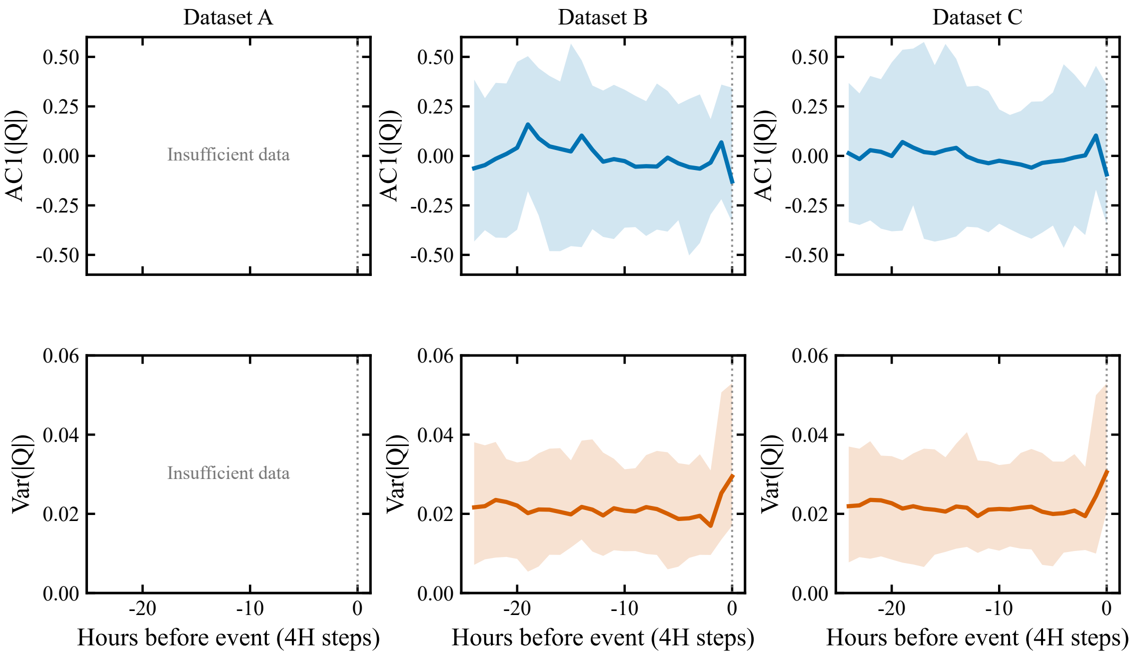


Figure S15: Event-aligned early-warning diagnostics under realistic missingness. We report event-aligned trajectories for autocorrelation and variance indicators under the primary 4-hour aggregation and illustrate how higher-frequency aggregation can become infeasible due to continuity requirements.

## S6.4 H2 density diagnostics: 4H versus 12H aggregation

The media-dominance–volatility association is sensitive to sampling density and missingness. Under the primary 4-hour aggregation, the raw correlation is positive but largely attenuates after controlling for segment sampling density (number of valid windows per week). Aggregating to 12-hour windows increases per-window sample size at the cost of temporal granularity and yields more stable within-segment estimates of $r_{\text{proxy}}$ and volatility; under this aggregation, the density-controlled association remains significant.

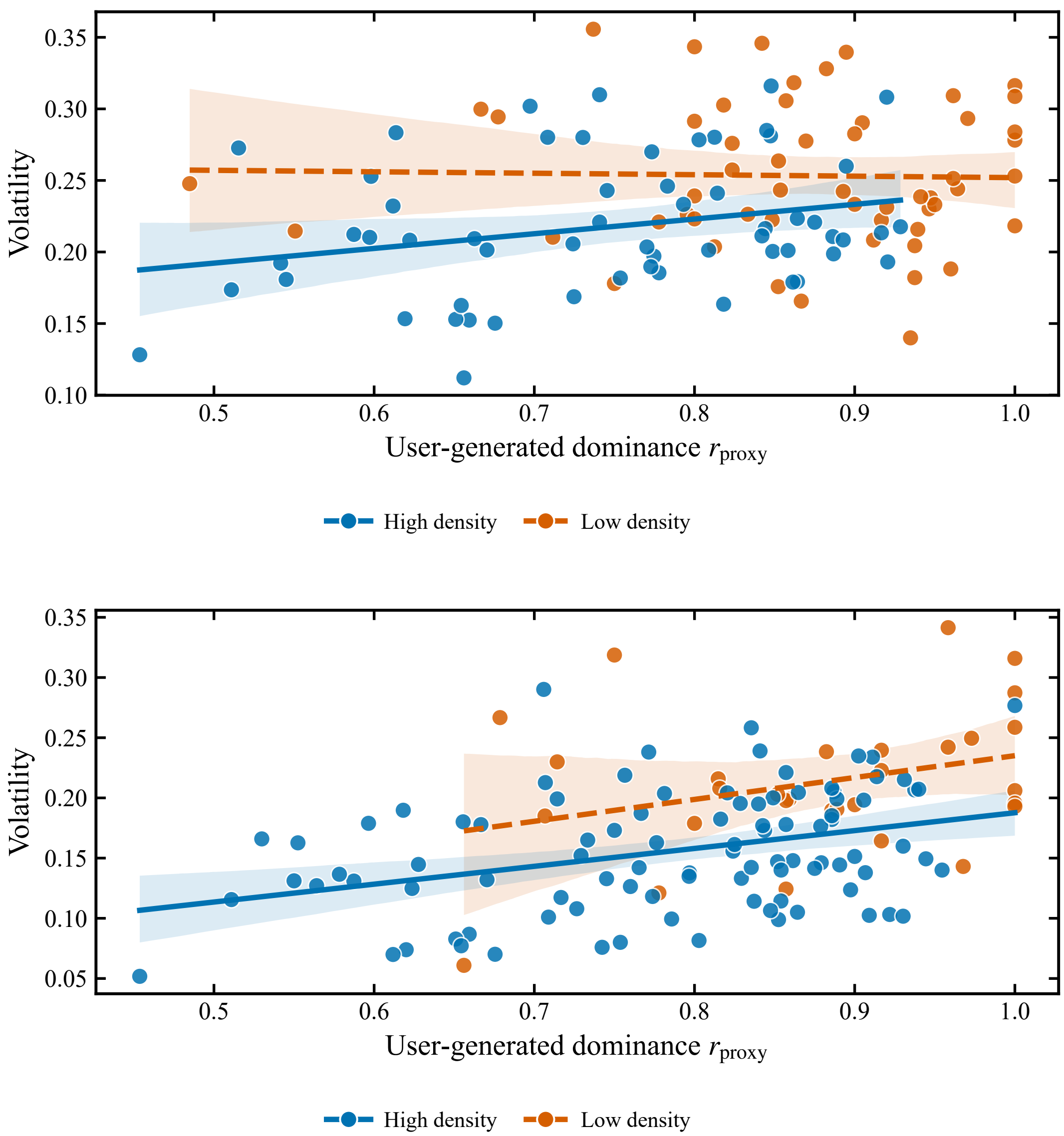


Figure S16: Density diagnostics for the media-dominance–volatility signature under two aggregation scales (high-density subset). **Top:** 4-hour aggregation (Pearson $r = 0.265$, $p = 0.00434$) attenuates after controlling for segment sampling density (partial $r = 0.078$, $p = 0.411$). **Bottom:** 12-hour aggregation yields a stronger association (Pearson $r = 0.429$, $p = 6.03 \times 10^{-7}$) that remains significant after density control (partial $r = 0.386$, $p = 8.79 \times 10^{-6}$). Solid/dashed lines show groupwise linear fits with 95% bootstrap bands.

# References

[1] Qwen Team. Qwen3-8b. Hugging Face model card, 2025. Accessed 2026-01-03.